FlowCode in Honeychrome

A walkthrough of every stage, with the outputs of a real screen

This guide follows one experiment through the six sections of the FlowCode tab, from choosing the backbone to the phenotype shift of each guide. Every picture is the app's own output on that experiment. Each stage says what the step does, shows what it produced, and lists what to look for before moving on. For the settings and the mathematics behind each step, see the plugin README; for installing and updating, the install guide.

In this example

- **JA_007 Metab Library 1 and 2.** Two metabolism CRISPR libraries of 56 guides each, read out with a FlowCode panel of 8 epitope tags (HA, Ollas, C, VSVg, S, V5, AU1, Flag) on a full-spectrum panel of T-cell markers.
- 145 samples: 11 tissues (spleen, axillary and mesenteric lymph nodes, Peyer's patches, liver, and the intraepithelial and lamina propria fractions of small intestine, caecum and colon) from 13 mice (mice 001–007 carry library 1, mice 008–013 library 2), plus 2 FRET backbone controls.
- Acquired over two days (4 and 5 July 2026), unmixed with the AutoSpectral noise model (M2pg).
- Analysed in three populations: CD4, Treg and CD8 T cells, each tissue against spleen, paired by mouse.

1	**Inputs**	backbone, matched unstained, AF library
2	**Tags & Combos**	tag channels, combinations, libraries
3	**Gates & Thresholds**	FlowCode scatter gates, one positivity threshold per tag
4	**Debarcode & FRET**	barcodes and the FRET/spillover error of each combination
5	**Unmix & Export**	corrected FCS files
6	**Analysis**	gating, counts, statistics, markers, pathways, gene groups, phenotype shift

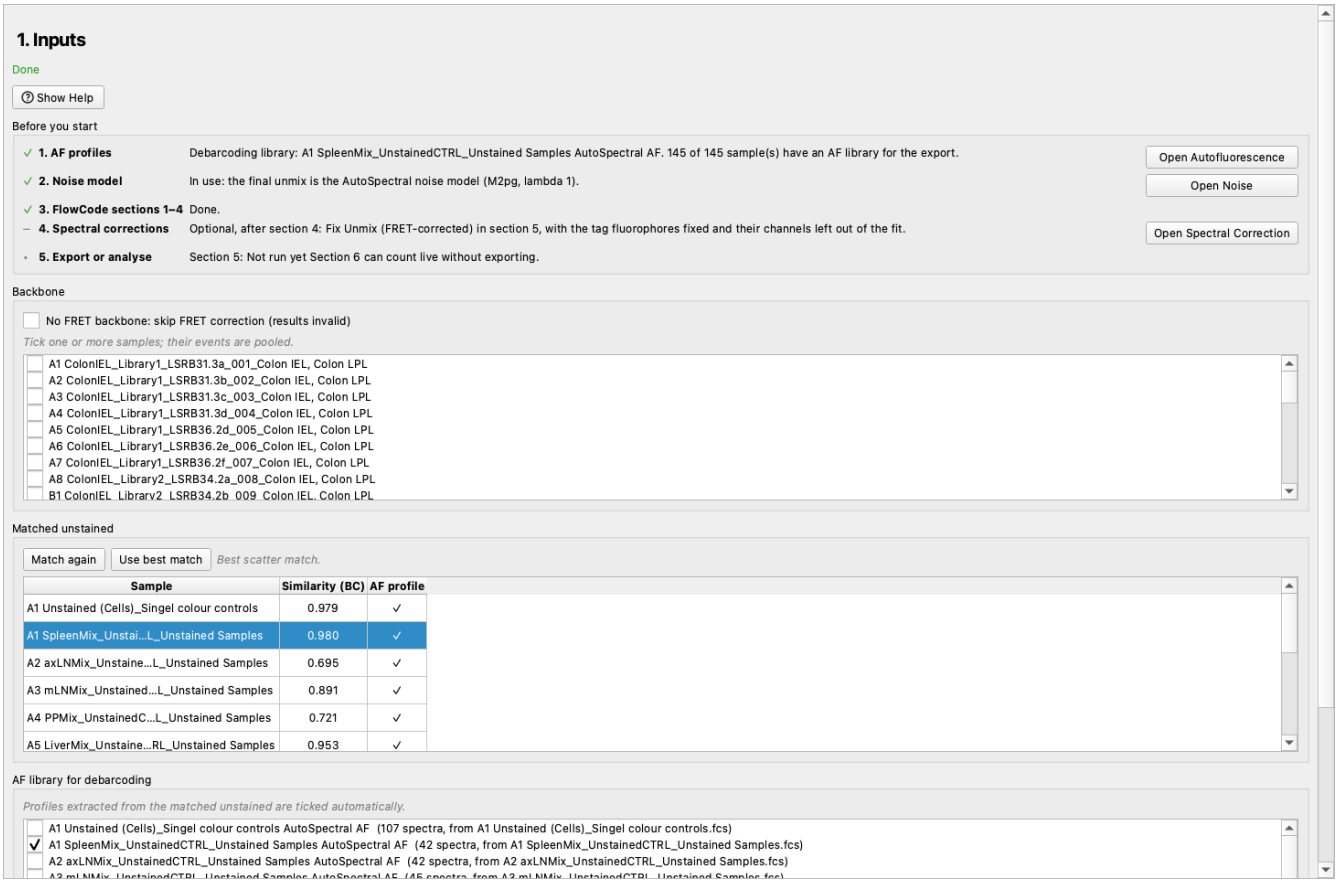
Boxes: **What good looks like** describes a healthy result; **What to check** lists the problems that step can hide.

Made on 8 November 2026 with Honeychrome 1.0.0 and FlowCode plugin 0.4.1. Screenshots come from the app run without a display, so windows are drawn at a fixed size.

SECTION 1

Inputs

The **backbone** is a pool of cells carrying every combination, acquired as its own sample (the FRET control). It is used to measure, once per experiment, how much each tag combination leaks into the other channels. The **matched unstained** sample is the unstained control whose scatter looks most like the backbone; its autofluorescence (AF) profiles are used while debarcoding. The *Before you start* box lists what must be done on the AutoSpectral tab first, with buttons that open each page.



Section 1 on JA_007. Every one of the 145 samples has an AF library, the final unmix is the noise model, and A1 SpleenMix unstained matches the backbone's scatter best (Bhattacharyya coefficient 0.980; 1 = identical).

What good looks like

- All items of *Before you start* ticked, except the optional spectral corrections.
- A matched unstained with a similarity above about 0.9, and an AF library of at least 2 spectra.

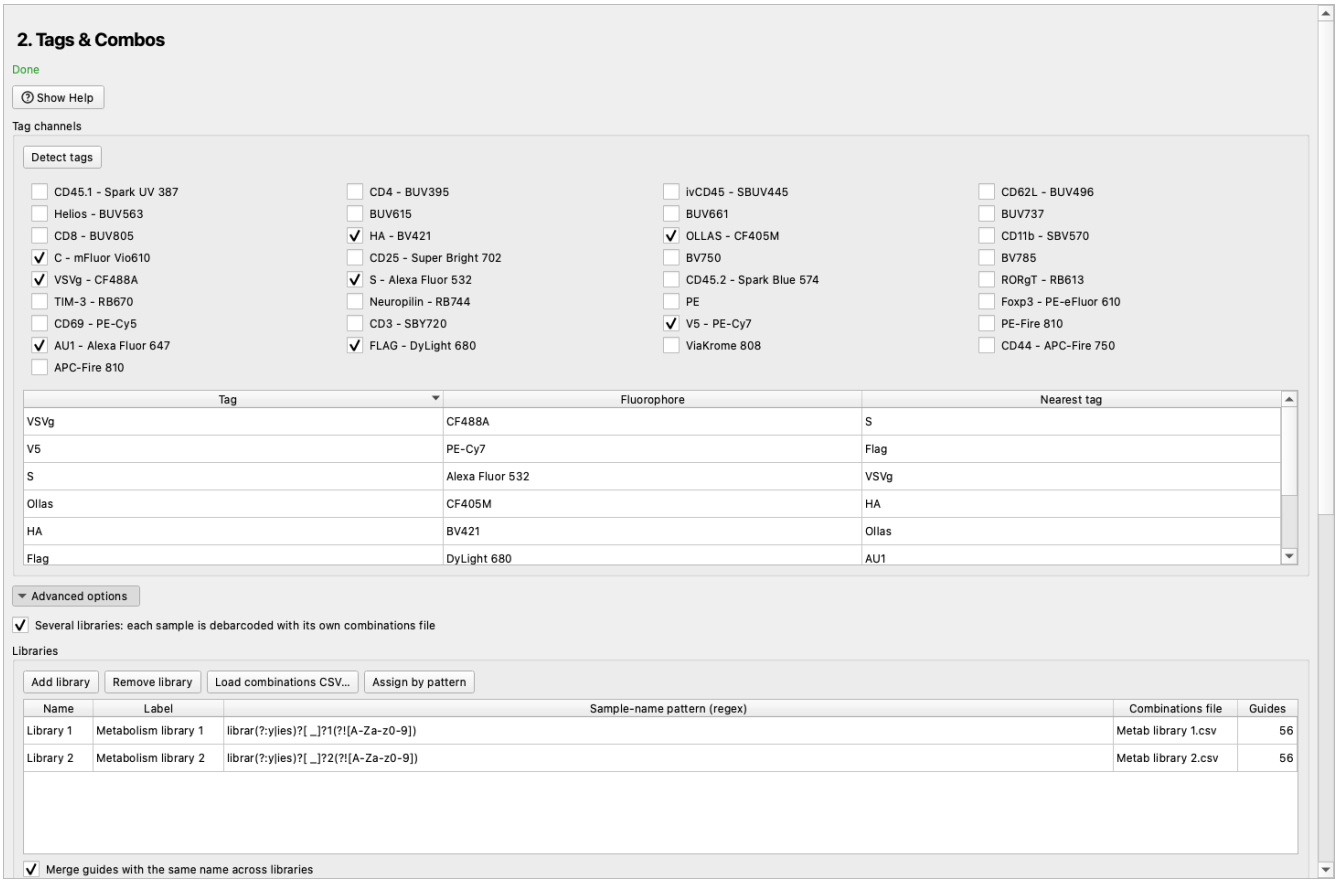
What to check

- Only the FRET backbone samples are ticked as backbone, not stained tissue samples.
- *No FRET backbone* stays unticked unless the experiment really has no FRET control: its results are marked invalid everywhere.

SECTION 2

Tags & Combos

The tag channels are the unmixed channels whose antigen is a FlowCode tag; they are ticked automatically. Each guide is a combination of three tags. JA_007 uses two libraries on the same tags, so *Several libraries* is ticked: each library has its own combinations CSV, and each sample is assigned to its library by a sample-name pattern (Library1, Library 2 and so on).



Eight tags are ticked (HA BV421, Ollas CF405M, C mFluor Vio610, VSVg CF488A, S Alexa Fluor 532, V5 PE-Cy7, AU1 Alexa Fluor 647, Flag DyLight 680). Library 1 and 2, labelled "Metabolism library 1/2", have 56 guides each.

What good looks like

- Exactly the tag channels are ticked, and no marker channel.
- Every library shows the number of guides you expect, and every sample has a library (FMOs and controls go to *Other*).

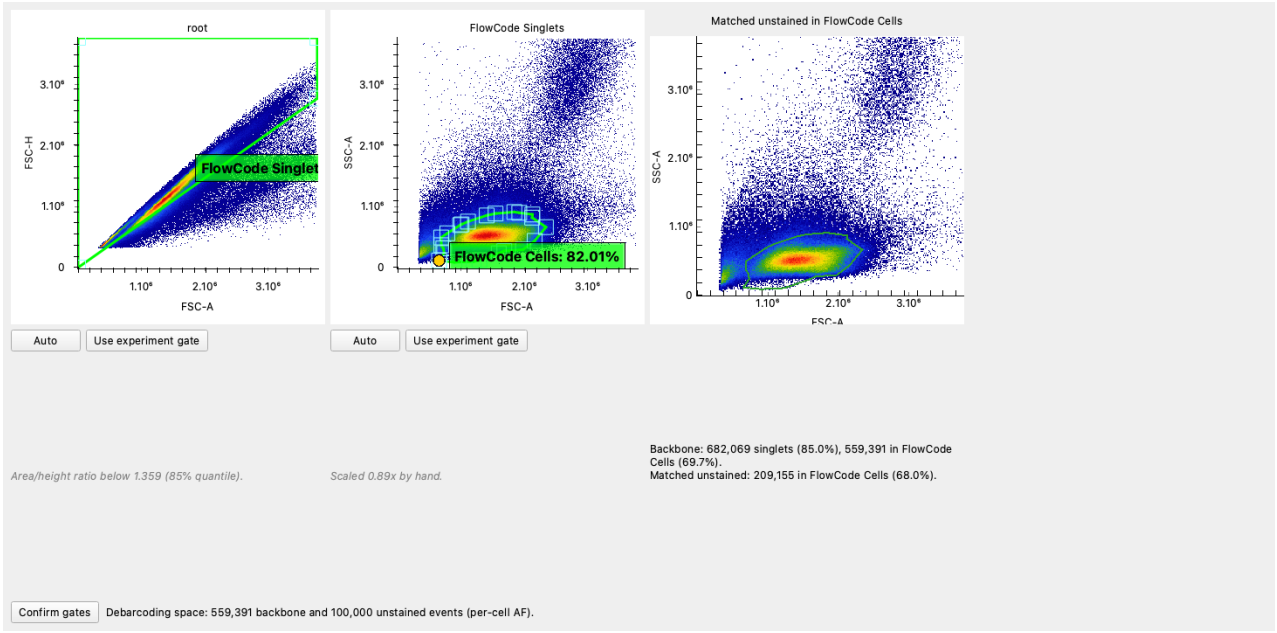
What to check

- *Nearest tag* pairs with very similar spectra (here V5 and Flag, Ollas and HA) need the most care at the threshold step.
- Guide names repeated across libraries (non-targeting controls) are merged only when *Merge guides with the same name* is ticked.

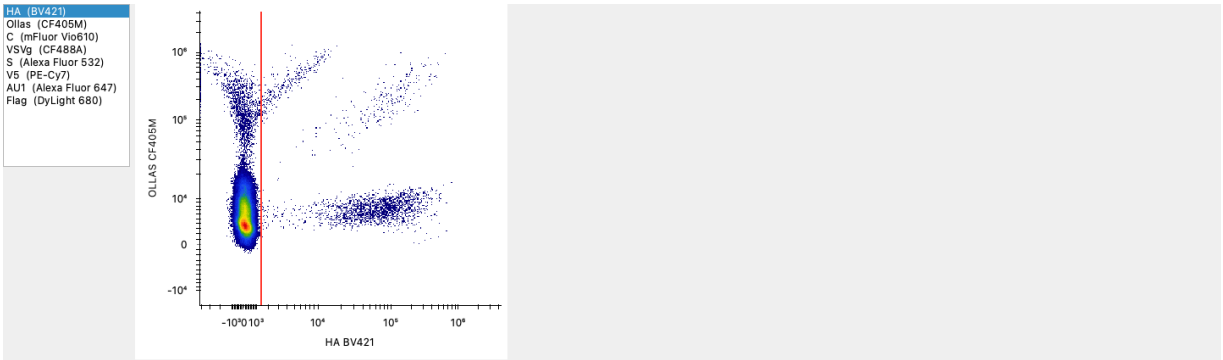
SECTION 3

Gates & Thresholds

Two scatter gates are drawn on the backbone: **FlowCode Singlets** (FSC area against height) and **FlowCode Cells**, a contour around the events most clearly positive for three tags. *Confirm gates* then unmixes the gated backbone in the debarcoding space and proposes one positivity threshold per tag. The threshold is the same for every sample.



Backbone: 682,069 singlets (85.0 %), 559,391 in FlowCode Cells (69.7 %); the matched unstained (right) shows the same Cells gate. The Cells gate was scaled to 0.89× by hand.



Auto-set all | Reset selected to auto | Import thresholds CSV...

Tag	Fluorophore	Y channel	Threshold (raw)	Threshold (logicle)	Method	ΔBIC
HA	BV421	CF405M	1417.6	0.203	manual	11392.3
Ollas	CF405M	BV421	24619.7	0.499	manual	2434.4
C	mFluor Vio610	Alexa Fluor 647	2320.5	0.248	manual	10058.1
VSVg	CF488A	Alexa Fluor 532	1908.1	0.229	manual	11220.5
S	Alexa Fluor 532	CF488A	3496.5	0.291	manual	8637.8
V5	PE-Cy7	DyLight 680	1928.3	0.230	mixture	11908.8

Threshold plot for HA against its nearest tag, Ollas. The red line is the threshold (1,417.6 raw units); drag it, or edit the raw value in the table.

What good looks like

- Each threshold sits in the gap between the negative cloud and the tag-positive arms.
- Positive arms are straight; the negative cloud is not cut through.

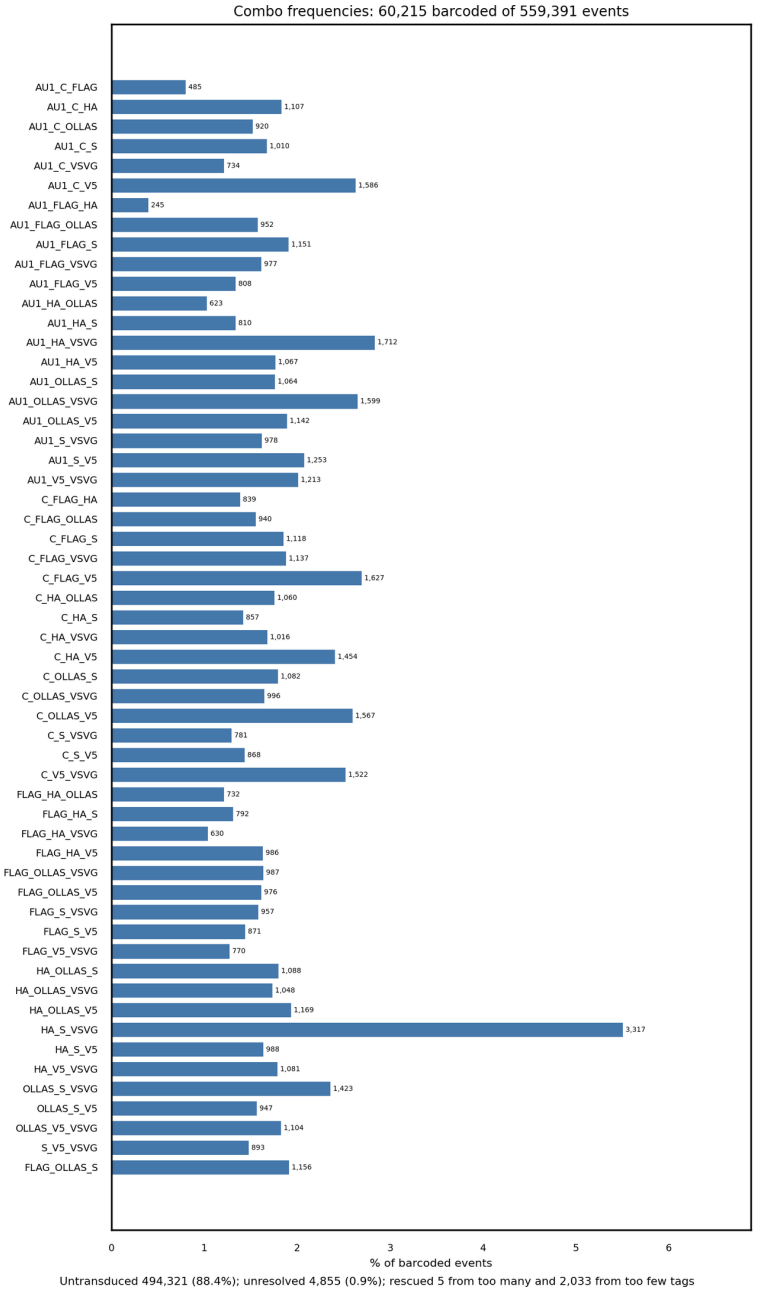
What to check

- Bright single-positive events leaning into the neighbouring tag: that is spillover, and section 4 corrects it, so set the threshold on the main gap, not on the lean.
- Editing a threshold makes sections 4 and 5 need a re-run.

SECTION 4

Debarcode & FRET

Debarcode and identify FRET spillover assigns each backbone event to a combination, rescues events with too many or too few positive tags, and then measures, for every combination, the extra signal it adds to the other channels. Tags sitting on the same cell transfer energy to each other (FRET), so this error grows with the product of the tags' brightness and differs from combination to combination. It is fitted on the bright, cleanly barcoded events and removed from the raw data before the final unmix.



Combination frequencies on the backbone: 60,215 barcoded of 559,391 events. All 56 combinations are present (245 to 3,317 events). Below the chart: untransduced 88.4 %, unresolved 0.9 %, 5 rescued from too many tags and 2,033 from too few.

What good looks like

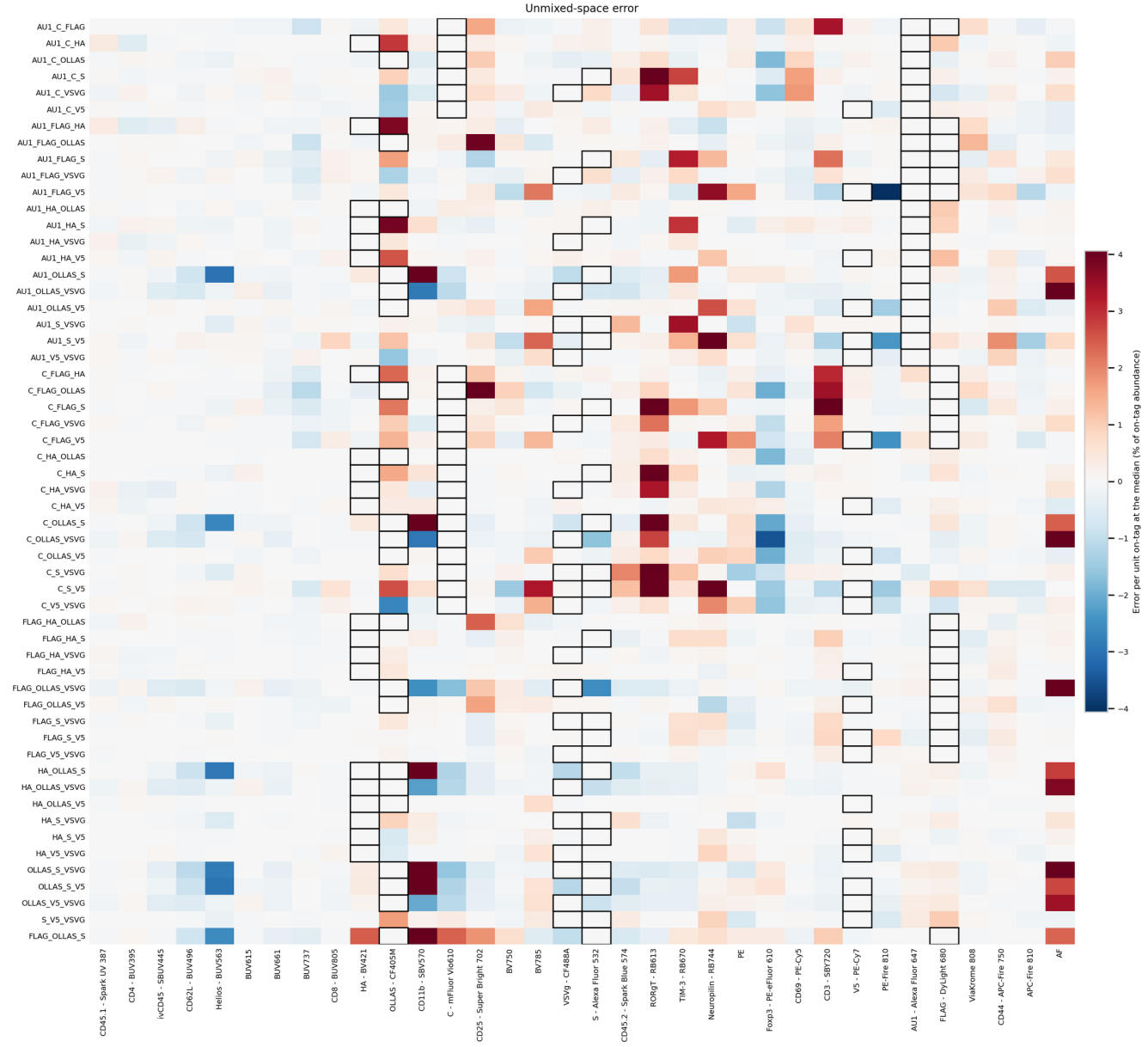
- Every combination present with at least about 50 bright events, so it gets its own error model.
- Unresolved events a small share (here 0.9 %).

What to check

- A combination far above the others (here HA_S_VSVG) or missing: check the library and the thresholds of its tags.
- Many events rescued from too few tags: a threshold may be set too high.

FRET and spillover error

Each row is a combination, each column a channel. The colour is how much that channel gains per unit of the combination's own tags, at the combination's median brightness, in percent. The combination's own tags are outlined.



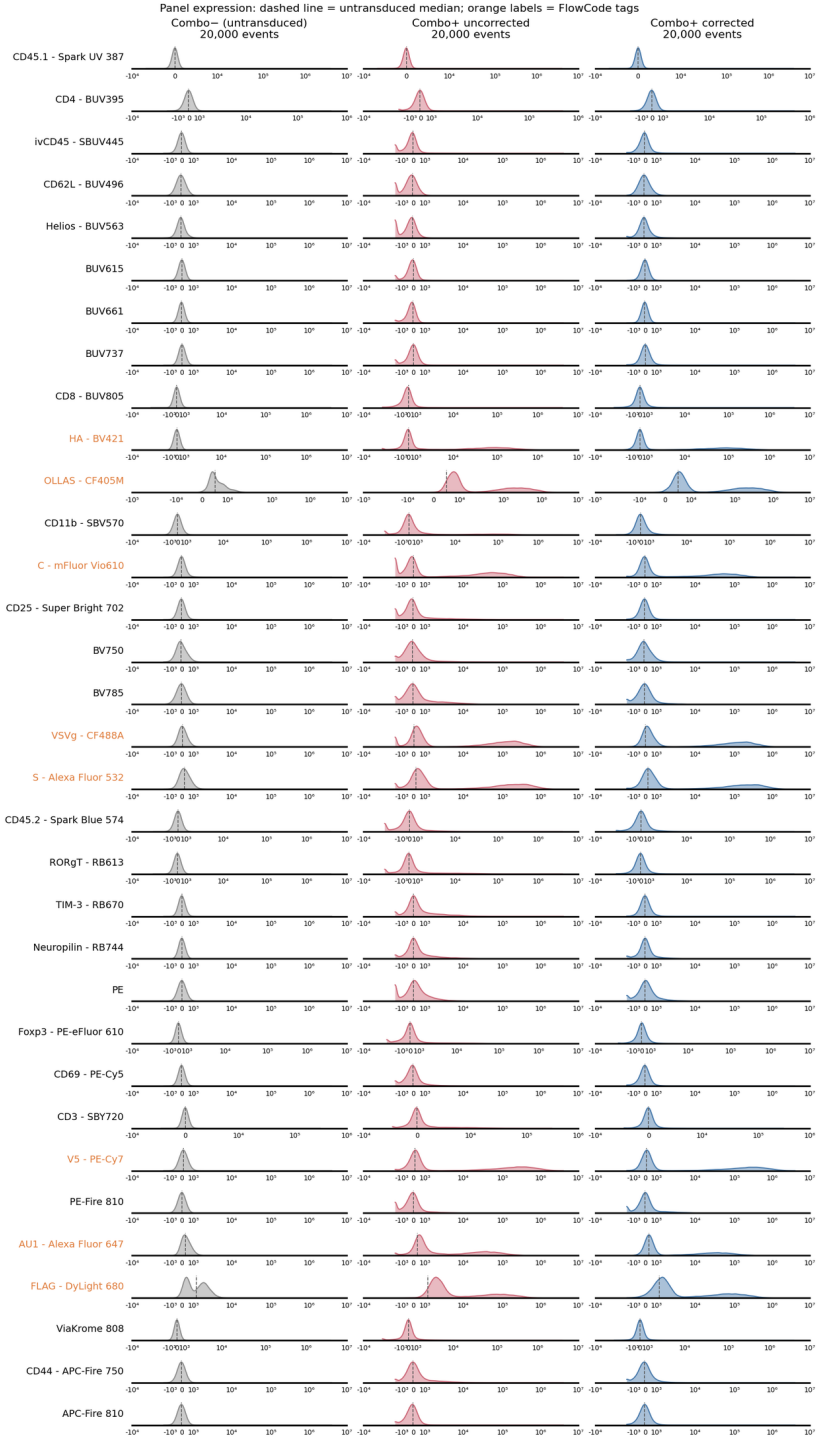
Error heatmap of JA_007. Strong errors appear in specific channels for specific tag pairs (for example SBV570 for combinations with Ollas and S, RORγT RB613 for those with C and S); this is what the correction removes.

What to check

- Large errors in a marker channel that you gate on (here RB613, PE-eFluor 610 or Super Bright 702): look at that marker in *Panel expression* and *Assess correction*.
- A whole column coloured for every combination with one tag: that tag's reference spectrum may be wrong, which a FRET correction should not be left to absorb.

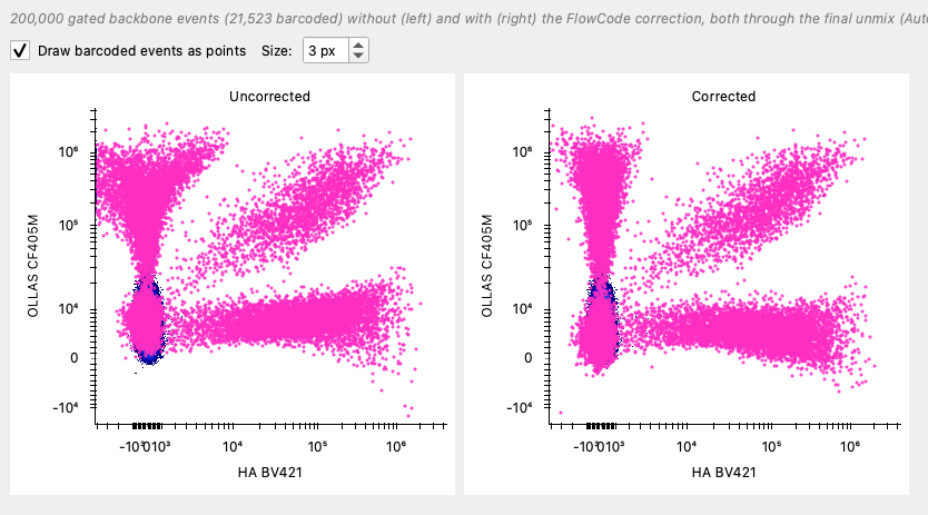
Panel expression before and after correction

One row per channel: untransduced cells (grey), barcoded cells without the correction (red) and with it (blue), all through the final unmix. Tag channels are labelled in orange.



After correction (right), the barcoded cells' negative peaks line up with the untransduced cells (dashed line) in the marker channels; before it (middle), several show shoulders and negative tails.

Assess correction and fit diagnostics



Ollas against HA on 200,000 backbone events (21,523 barcoded, pink), before and after the correction, both through the final unmix. Click an axis label to compare any other pair of channels.

Kappa: condition number of the on-tag fit. In/Out of span: error inside and outside the space the reference spectra can unmix, relative to the on-tag spectra. Off RMS/spill: median off-target abundance and largest off-target slope before and after correction, on the fit events and on held-out (rescued) events.

Combo	Tags	Error	Confident	Fit	Kappa	In span	Out of span	Curvature	AF slope	AF excess	AF confound	Off RMS before	Off RMS after	Off spill before	Off s
S_V5_VSVG	S VSVg V5	combo	891	835	6.177e+08	0.09089	0.02295	0.1787	-0.000393	-0.001248	0.007434	923	115.4	0.03481	0.0300
OLLAS_V5_VSVG	Ollas VSVg V5	combo	1064	954	2.302e+08	0.06138	0.01667	0.0758	0.02673	0.02587	0.381	2627	60.18	0.06154	0.0124
OLLAS_S_VSVG	Ollas S VSVg	combo	1420	1308	1.982e+08	0.1863	0.0572	0.01262	0.04922	0.04836	0.282	15300	84.15	0.2272	0.0044
OLLAS_S_V5	Ollas S V5	combo	945	858	1.751e+08	0.1613	0.05284	0.02707	0.02473	0.02387	0.1306	14000	138.1	0.2245	0.0108
HA_V5_VSVG	HA VSVg V5	combo	1078	954	2.384e+07	0.03614	0.01522	0.3721	0.001675	0.0008193	0.06129	548.6	85.83	0.04016	0.0066
HA_S_VSVG	HA S VSVg	combo	3316	3204	5.246e+07	0.046	0.004661	0.1515	0.00291	0.002054	0.08576	695.6	91.82	0.03646	0.0067
HA_S_V5	HA S V5	combo	981	869	5.144e+07	0.02994	0.005628	0.2488	0.001158	0.0003024	0.06076	640	106.6	0.02813	0.0054
HA_OLLAS_VSVG	HA Ollas VSVg	combo	1011	786	8.599e+07	0.05196	0.01386	0.05421	0.04199	0.04113	0.6148	1825	37.08	0.06071	0.0009
HA_OLLAS_V5	HA Ollas V5	combo	1153	1018	1.55e+08	0.0126	0.002507	0.1581	0.0002178	-0.0006375	0.01888	880.4	57.71	0.01876	0.0017
HA_OLLAS_S	HA Ollas S	combo	1055	910	1.289e+08	0.141	0.04544	0.02154	0.0318	0.03095	0.1784	10250	131.4	0.2139	0.0043
FLAG_V5_VSVG	Flag VSVg V5	combo	648	331	3.142e+09	0.01925	0.01459	0.4098	0.0003363	-0.000519	0.03536	822.1	85.86	0.02734	0.0039
FLAG_S_VSVG	Flag S VSVg	combo	879	646	2.359e+08	0.03272	0.01358	0.08107	0.000597	-0.0002583	0.01898	2661	56.23	0.02843	0.0031
FLAG_S_V5	Flag S V5	combo	857	615	1.587e+09	0.02993	0.01533	0.1213	0.0006361	-0.0002192	0.01799	2283	234.1	0.02238	0.0053
FLAG_OLLAS_VSVG	Flag Ollas VSVg	combo	850	574	4.308e+07	0.06499	0.01469	0.1019	0.04292	0.04207	0.5084	3504	40.85	0.07383	0.0267
FLAG_OLLAS_V5	Flag Ollas V5	combo	838	664	1.441e+08	0.04358	0.02323	0.1101	2.974e-05	-0.0008256	0.0005139	1836	56.38	0.0713	0.0158
FLAG_OLLAS_S	Flag Ollas S	combo	1087	850	3.917e+07	0.1785	0.04834	0.03584	0.03135	0.03049	0.1384	15880	120	0.2115	0.0631
FLAG_HA_VSVG	Flag HA VSVg	combo	507	245	9.813e+08	0.02146	0.005917	0.2348	0.00398	0.003125	0.3709	773.9	66.96	0.01755	0.0031
FLAG_HA_V5	Flag HA V5	combo	852	601	4.134e+09	0.02085	0.009113	0.3491	0.0004696	-0.0003857	0.03161	955.8	190.5	0.0213	0.0069
FLAG_HA_S	Flag HA S	combo	746	555	5.995e+08	0.02415	0.01246	0.0736	0.003382	0.002526	0.09887	2140	90.02	0.02493	0.0033
FLAG_HA_OLLAS	Flag HA Ollas	combo	587	445	6.522e+07	0.03436	0.006227	0.08459	0.0007404	-0.0001149	0.01327	1658	46.32	0.08328	0.0087
C_V5_VSVG	C VSVg V5	combo	1514	1379	3.673e+09	0.09755	0.04554	0.2373	0.0005751	-0.0002802	0.00664	493	68.69	0.07363	0.0685
C_S_VSVG	C S VSVg	combo	769	689	1.602e+08	0.2362	0.01174	0.05552	-0.0004678	-0.001323	0.002068	2387	63.43	0.3968	0.0307
C_S_V5	C S V5	combo	853	725	2.289e+09	0.2651	0.118	0.1129	5.466e-05	-0.0008006	0.0001937	2076	115.6	0.353	0.0303
C_OLLAS_VSVG	C Ollas VSVg	combo	961	727	2.875e+07	0.0939	0.02179	0.06973	0.04337	0.04251	0.3837	1928	47.49	0.1029	0.0098
C_OLLAS_V5	C Ollas V5	combo	1560	1347	4.677e+07	0.04883	0.02089	0.1469	-0.0001077	-0.000963	0.00201	849.1	48.16	0.06709	0.0125
C_OLLAS_S	C Ollas S	combo	1052	871	1.065e+07	0.1936	0.05236	0.02401	0.03502	0.03416	0.1705	10550	102	0.225	0.0077
C_HA_VSVG	C HA VSVg	combo	1014	955	1.638e+09	0.05557	0.01067	0.1446	0.004018	0.003163	0.07114	1061	115.2	0.1846	0.0507
C_HA_V5	C HA V5	combo	1447	1334	8.806e+08	0.02474	0.01234	0.3144	0.0006838	-0.0001715	0.03061	612.8	106.4	0.03451	0.0196

Fit diagnostics: per combination, the events used for the fit, the conditioning (kappa), the error inside and outside the reference spectra's span, its curvature, and the off-target signal before and after correction.

What good looks like

- Off-tag channels of barcoded and untransduced cells line up after correction.
- *Off RMS after* far below *Off RMS before* for every combination.

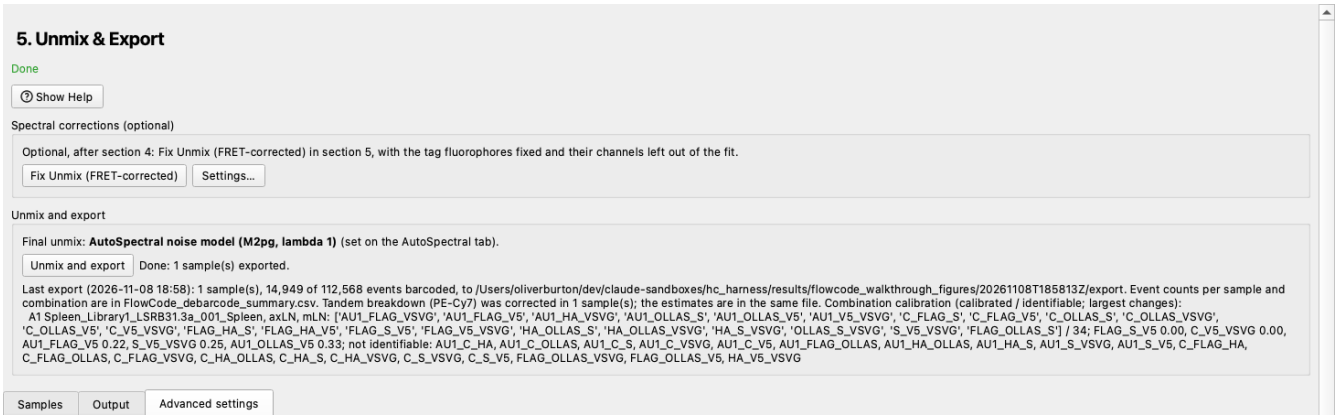
What to check

- Few *Fit* events: such combinations fall back to the shared pairwise model.
- Re-open the saved run with **Show event figures**; it never re-runs the fit.

SECTION 5

Unmix & Export

Each selected sample is debarcoded with the backbone's thresholds, the characterised error is subtracted from its raw data, and it is unmixed with the experiment's method. Two per-sample checks run on the way: the **tandem breakdown** of tags on tandem dyes (V5 on PE-Cy7 leaking into PE) and the **combination calibration**, which rescales each combination's correction to what the sample itself shows. Section 6 can also do all of this live, without writing files.



One spleen sample exported. The result line reports the tandem estimate and how many combinations were calibrated.

Spleen sample A1 (library 1)	
Total events	112,568
Untransduced	89,107
Unresolved	8,512
Barcoded	14,949
Rescued: too many / too few tags	38 / 505
Corrected by the pairwise model	17,037
V5 (PE-Cy7) → PE breakdown	z = 20.7, above 3: corrected
Combinations calibrated	22 of 34 identifiable

From FlowCode_debarcode_summary.csv, written beside the exported files.

What an exported file contains

A1 Spleen_Library1_... FlowCode (Unmixed).fcs has the usual unmixed channels, AF Abundance and AF Index, then FlowCode (a barcoded event's median tag abundance, 0 otherwise), FlowCode ID (0 untransduced, 1–56 the combination, 57 unresolved) and one FC <triplet> channel per combination; gate on FC AU1_C_FLAG > 0 to select that combination. With several libraries the channels are named by triplet, and the keyword FLOWCODEGUIDES gives each its guide in this sample's library (FC AU1_C_FLAG = Nrip1). Other keywords record the source sample, the library, the characterisation identifier and the calibration, so section 6 can check each file is current.

What good looks like

- Barcoded events a stable share across samples of a tissue.
- Tandem z below 3 in most samples; where it is above, the breakdown is corrected.

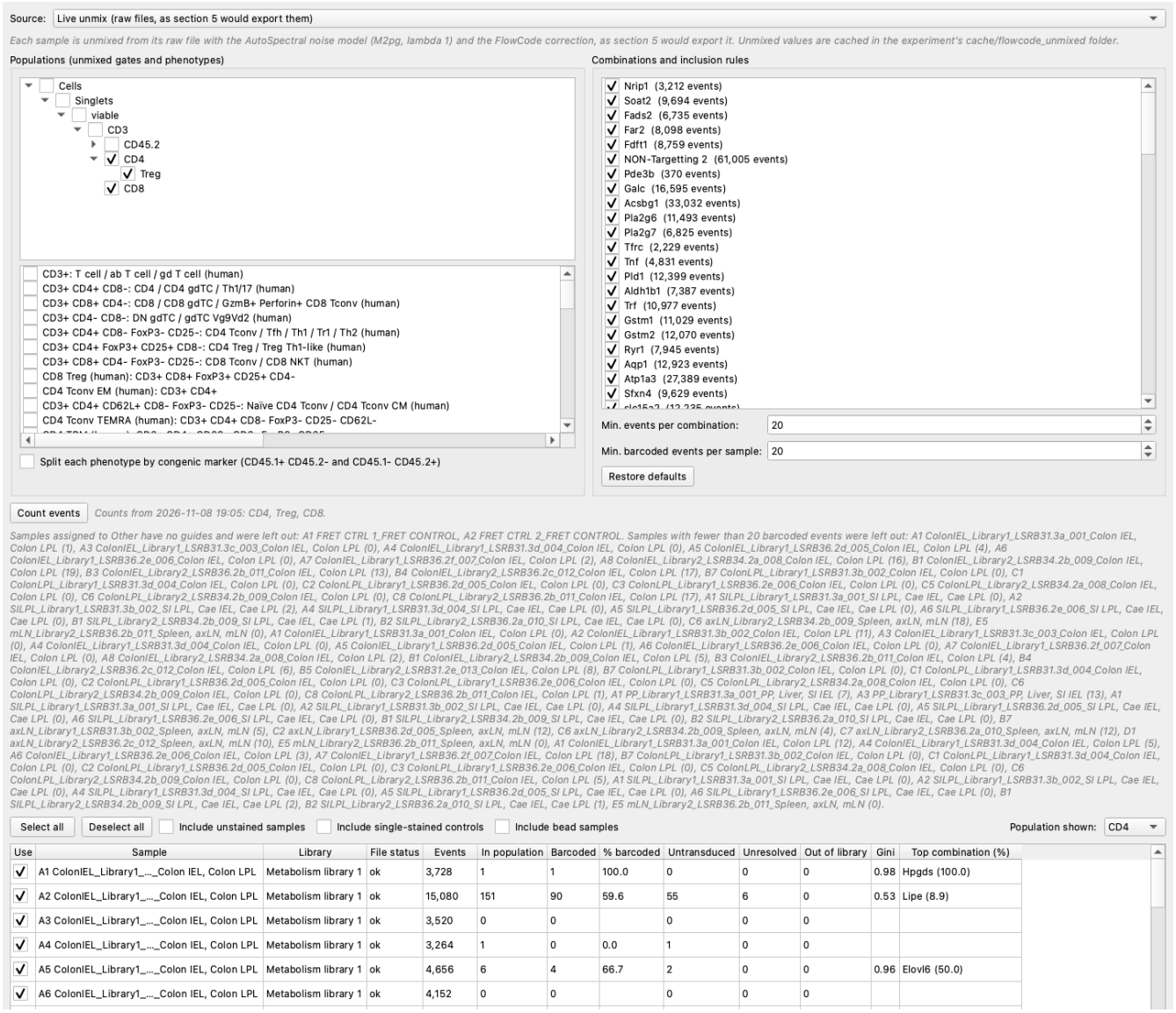
What to check

- Samples greyed out: their AF library has fewer than 2 spectra.
- A red note about Fix My Unmix on a tag fluorophore: tag channels must stay fixed there.

SECTION 6

Analysis: data and groups

The Analysis page counts the barcoded cells of each guide in each population of each sample. **Live unmix** (default) unmixes the raw files exactly as section 5 would, without writing them; **Exported files** reads the files section 5 wrote, and can add the exports of other experiments of the same study (other acquisition days). Populations are gates of the unmixed gating tree, applied to the FlowCode-corrected events, or gate-free phenotypes.



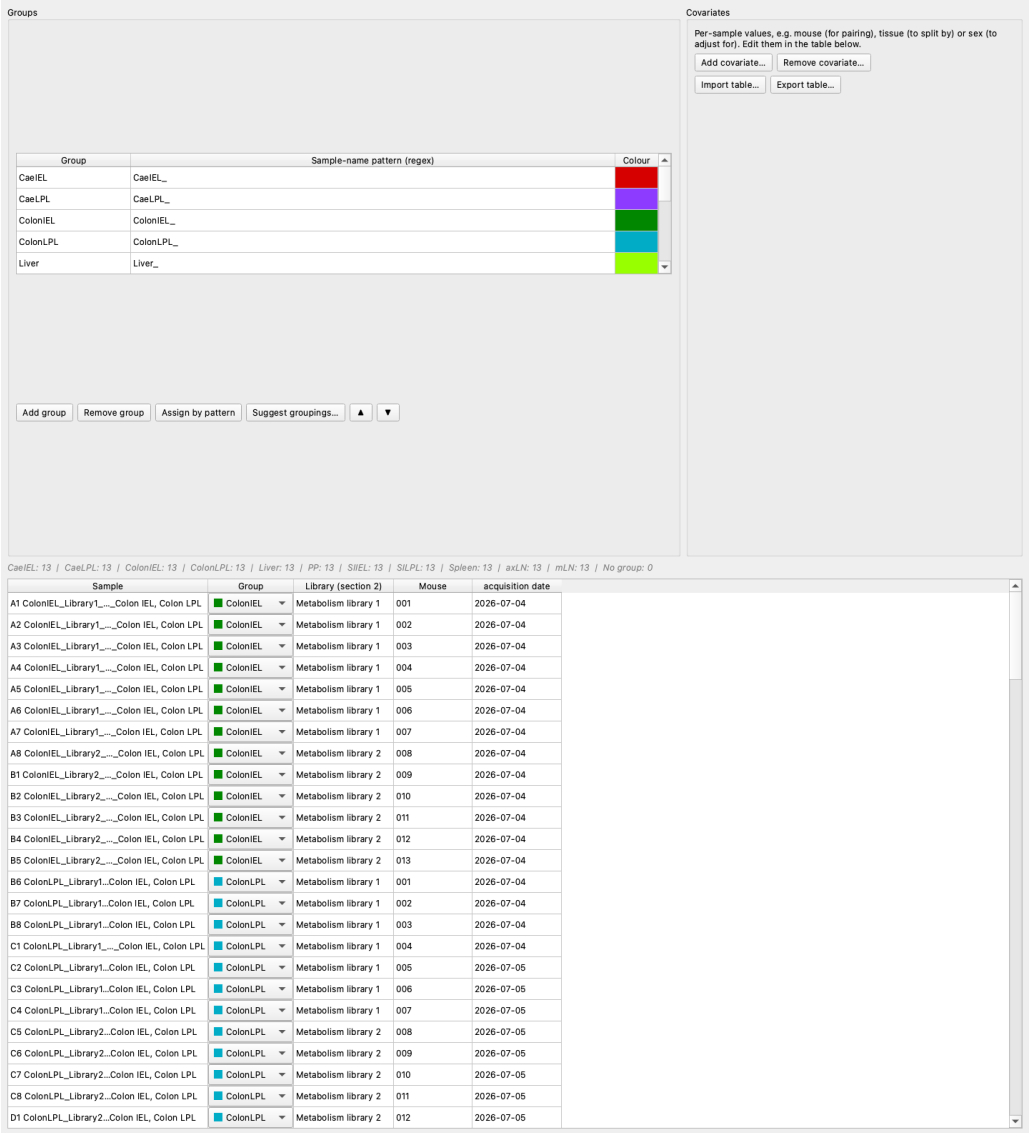
Data tab: CD4, Treg and CD8 ticked; 145 samples, 429 sample × population counts. The table shows each sample's events, barcoded share and its most frequent combination.

What to check

- Samples left out for too few barcoded events (listed above the table). In this experiment many colon IEL samples hold under 20 barcoded CD4 cells.
- A high *Gini* or a single combination above 50 %: an expanded clone, which the clone filter in Statistics can remove.

Groups, pairing and the acquisition date

Groups are set by sample-name patterns; each has a colour used in every figure. The covariates table holds the mouse (for pairing) and, because the files carry two acquisition dates, an *acquisition date* covariate that the page fills in and adjusts for on its own.



Eleven tissue groups of 13 samples each. Mouse 001–013 pairs the tissues of one animal; acquisition date 4 or 5 July comes from each file's $DATE.

In this example

- A mouse's tissues were acquired on both days here (for example mouse 005: colon IEL on 4 July, colon LPL on 5 July), so pairing by mouse does not cover the day and the date stays in the model. Had each mouse been acquired on one day, pairing alone would cover it.
- The same covariate is added when other days' experiments are linked under *Other experiments*.

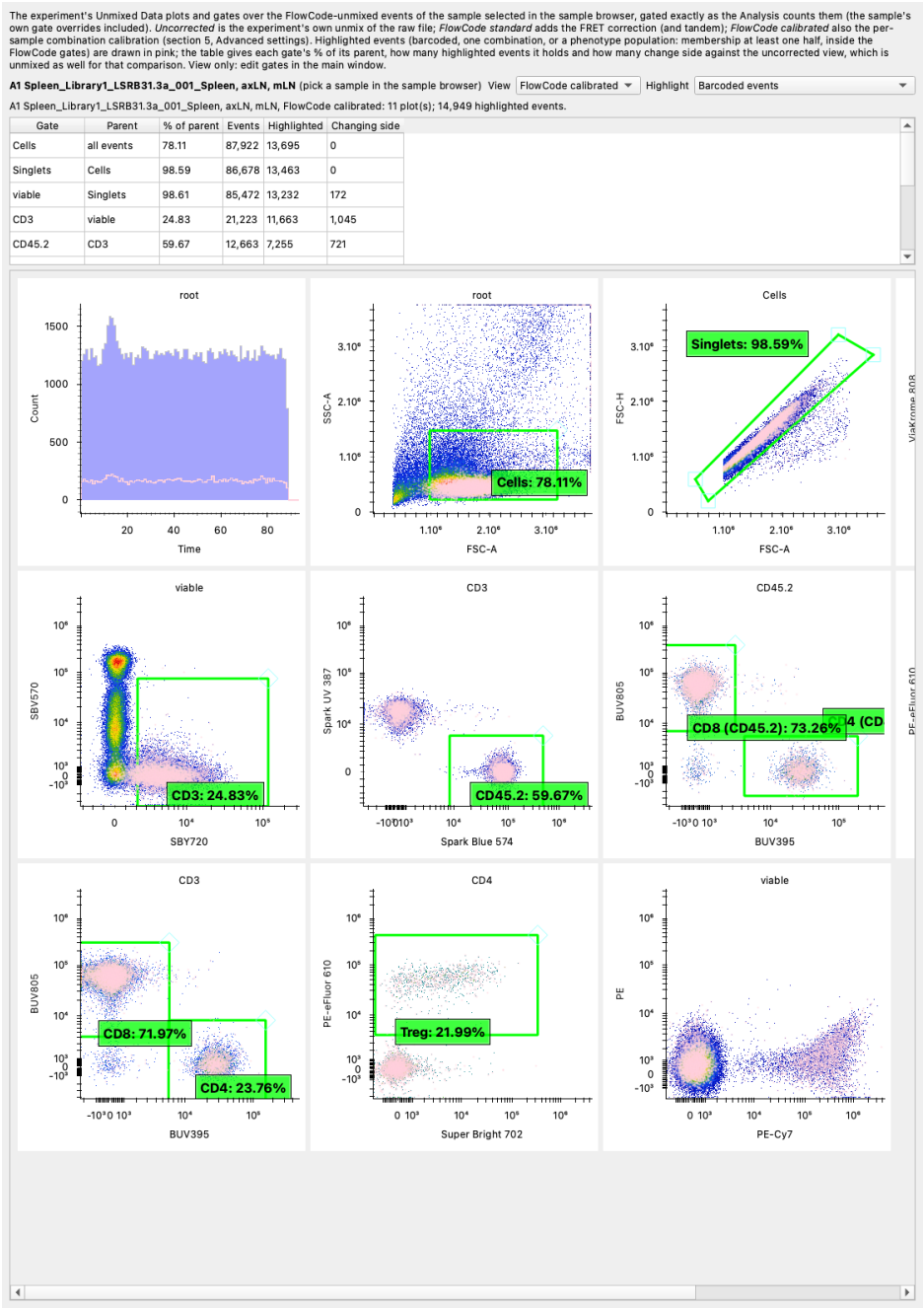
Statistics settings



Counts (quasi-likelihood F test), Frequencies and Markers ticked, robust normalisation, expanded clones removed, pathways and gene groups (the Metabolism table) on; each tissue against spleen within each population, paired by mouse, adjusted for acquisition date. The status line on the right reports what each test found and why samples were left out.

Gating on the corrected data

The Gating tab draws the experiment's unmixed plots and gates over the FlowCode-corrected events of the sample picked in the sample browser, gated exactly as the counts are. The table counts how many barcoded events change side of each gate against the uncorrected unmix.



Spleen A1, calibrated view, barcoded events in pink (14,949). 1,045 barcoded events change side of the CD3 gate between the uncorrected and corrected data, 172 of the viable gate.

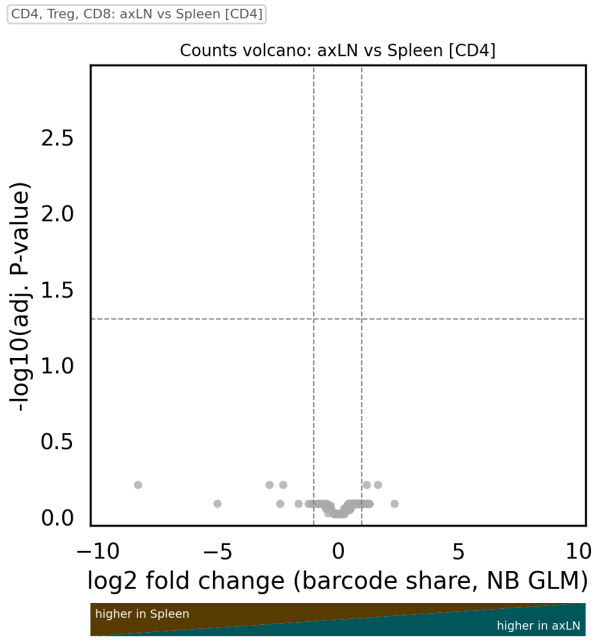
What to check

- Gates whose edge lies on a tag channel or a FRET-affected marker: barcoded events changing side there is the correction at work, but check the gate still fits the corrected data.

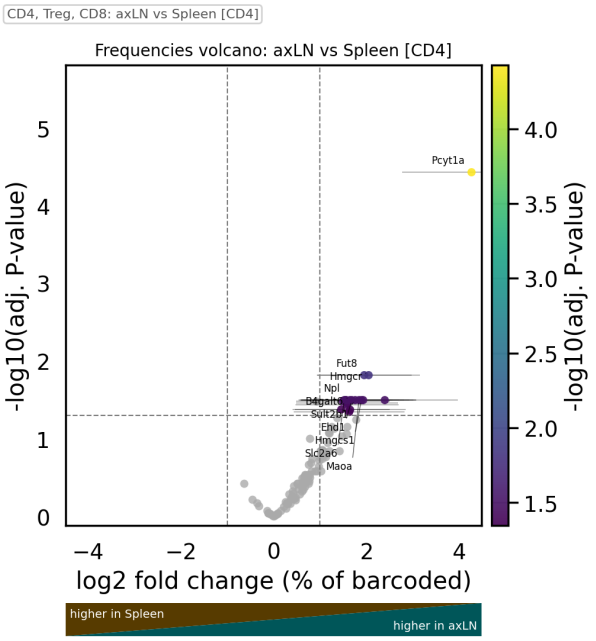
SECTION 6

Results: which guides change in number

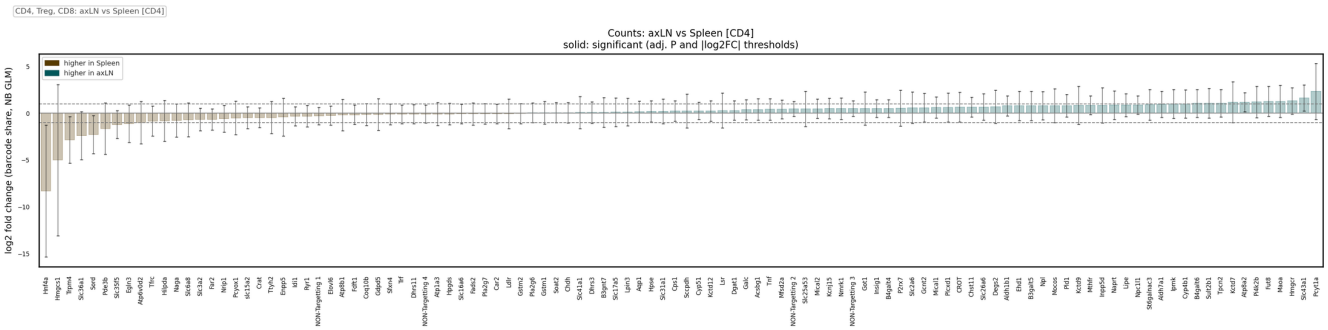
Each comparison asks whether a guide's share of the barcoded cells differs between two tissues. **Counts** fits a negative-binomial model to the cell numbers, weighting each sample by its cells; **Frequencies** fits a linear model to the log percentages. The two can disagree: the Counts test allows for extra mouse-to-mouse noise in small samples, so it is the more cautious.



Counts volcano, axillary LN against spleen, CD4: no guide passes.



Frequencies volcano, same comparison: 19 guides higher in axLN, led by Pcyt1a (log2 fold change 4.3).



Fold-change bars for every guide in the same comparison, with confidence intervals. Double-click a guide in a volcano, the bars or the table to plot it in every mouse.

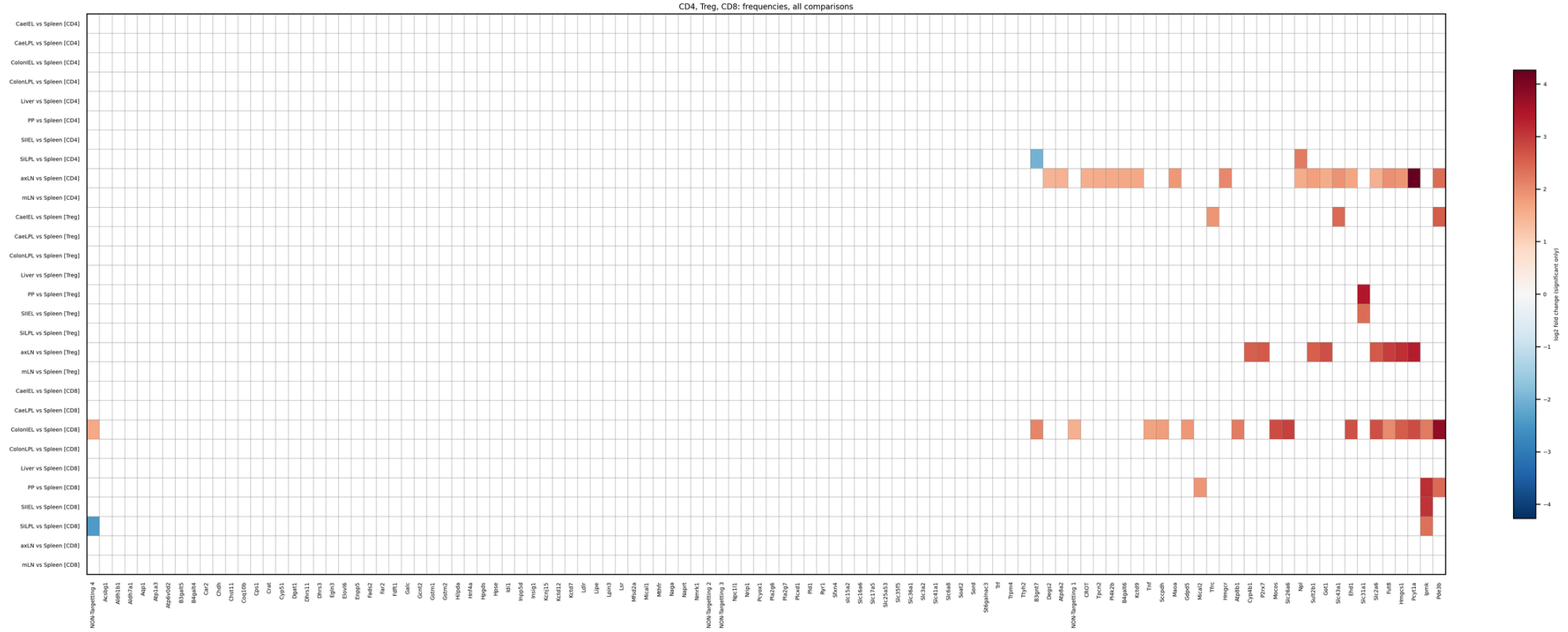
What good looks like

- Non-targeting guides near zero in every comparison.
- Hits that hold in both tests, and in *Check hits: leave one out*.

What to check

- Hits found by Frequencies only (as here): open the guide's per-mouse plot before trusting them.
- Wide confidence intervals mean few cells in some mice.

All comparisons at a glance



Comparison heatmap (Frequencies test): every comparison (rows) against every guide (columns), coloured only where significant. Axillary LN against spleen in CD4 and Treg, colon IEL against spleen in CD8, and Ipmk in the gut CD8 comparisons stand out. Two control guides are significant too: NON-Targetting 4 (first column) in colon IEL and SI LPL CD8, and NON-Targetting 1 in colon IEL CD8.

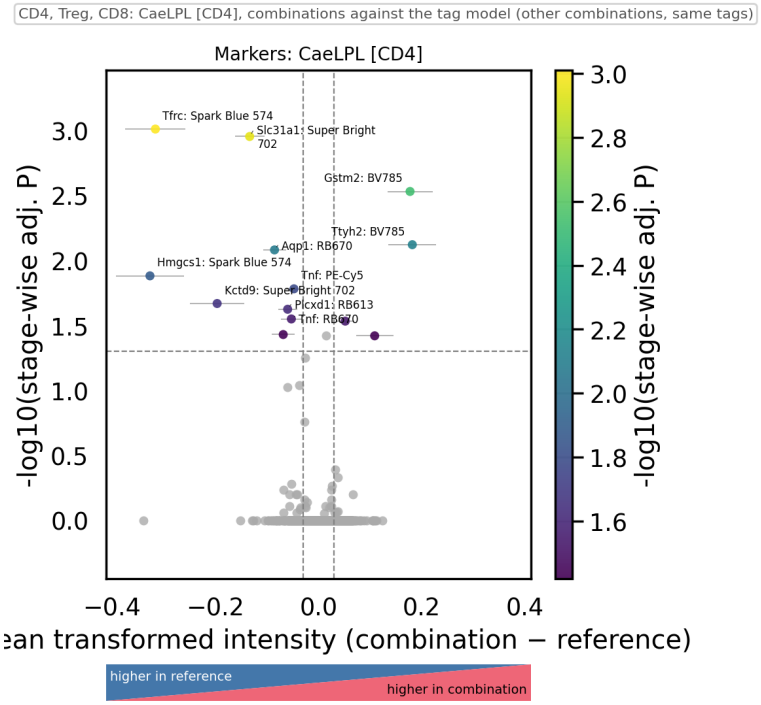
What to check

- A non-targeting guide among the hits is a warning that something other than the knockouts moves guide shares in that comparison (expanded clones, few cells, a gating difference). Look at its per-mouse plot and run *Check hits: leave one out* before reading the other hits of that comparison.

SECTION 6

Results: markers

The marker test asks whether a guide's cells are brighter or dimmer for a marker than its tags predict, from the other guides of the same mouse (the *tag model*). Tag spillover that is still in the data is added to each test's uncertainty, scaled by k, which is calibrated so that the non-targeting guides give no hits.



Marker volcano, caecum LPL CD4: each point is a guide and a marker. 13 pairs pass, for example Tfrc and Hmgcs1 knockouts lower on CD45.2 (Spark Blue 574) and Slc31a1 lower on CD25 (Super Bright 702). k was 0.25 in CD4 and Treg and 0.5 in CD8.

What good looks like

- Hits on markers the guide's biology can explain.
- Hatched cells in the heatmap are few: little of the uncertainty is spillover.

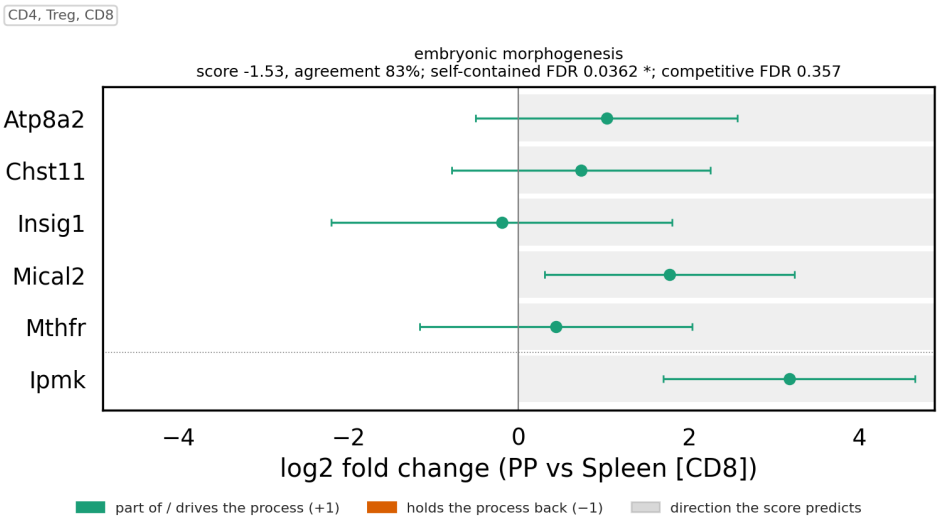
What to check

- Blocks of hits sharing one tag fluorophore's neighbour channel: leftover spillover, not biology.
- Congenic markers (CD45.1, CD45.2) differ between donor cells for reasons other than the guide.

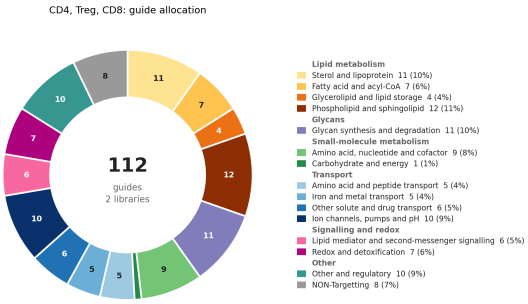
SECTION 6

Results: pathways and gene groups

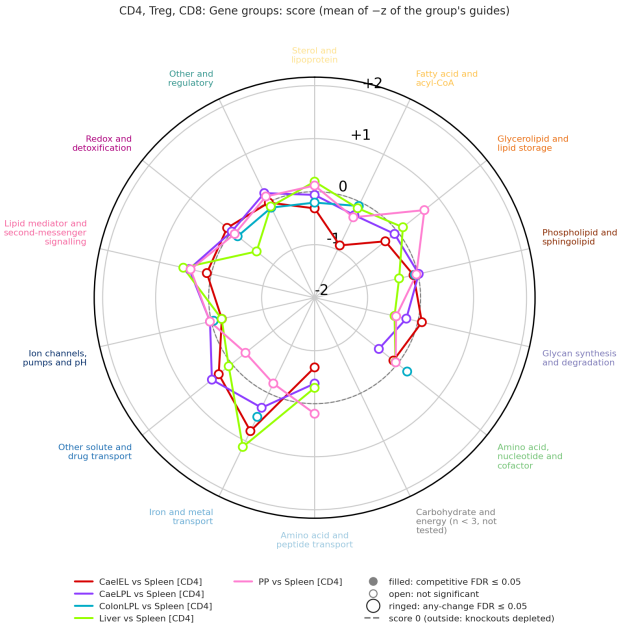
Single guides often move a little; a biological process or a family of genes taken together can move clearly. **Pathways** use Gene Ontology processes, where a knockout of a gene that drives the process and a knockout of one that holds it back should move in opposite directions. **Gene groups** use the lab's own table (here Metabolism: sterol, fatty acid, glycan metabolism, transporters and so on). Both are built on the Counts results.



Pathway members, Peyer's patches against spleen, CD8: "embryonic morphogenesis", score −1.53, agreement 83 %. The self-contained test passes (FDR 0.036) but the competitive test does not (0.357): the set moves, but not more than the rest of the library, and much of it is Ipmk, the strongest single guide.



Guide allocation: the 112 guides of both libraries by gene group.



Gene-group spider, CD4: one line per tissue against spleen; outside the dashed ring the family's knockouts are depleted. No group passes the competitive test in this run.

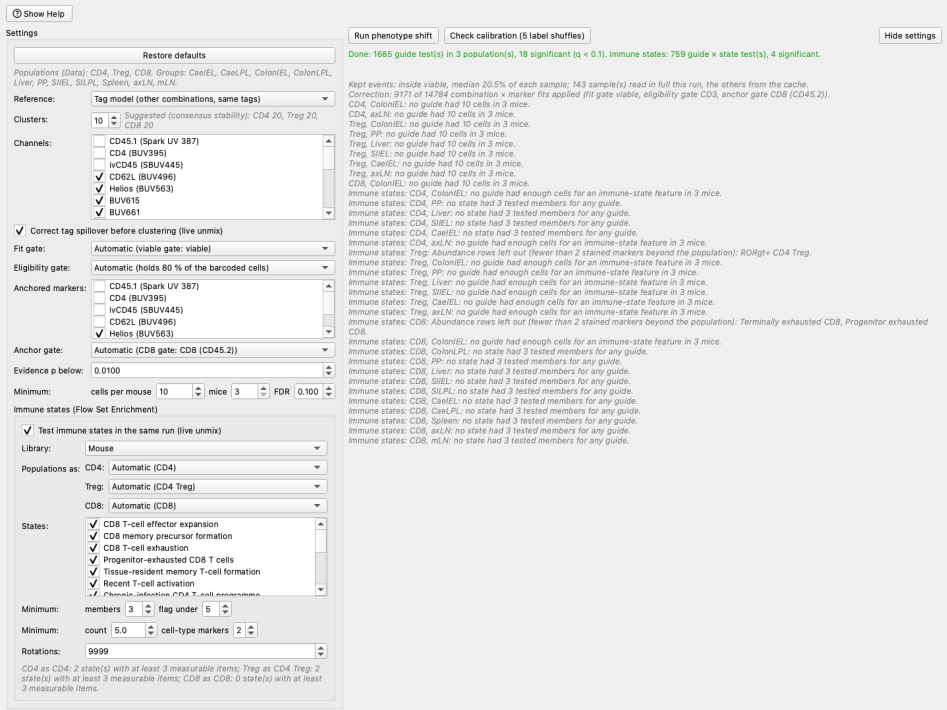
What to check

- Read the score, the agreement and the competitive FDR together: a set carried by one guide passes the self-contained test but is not a pathway effect.
- Groups with fewer than 3 tested guides are not tested (grey labels).

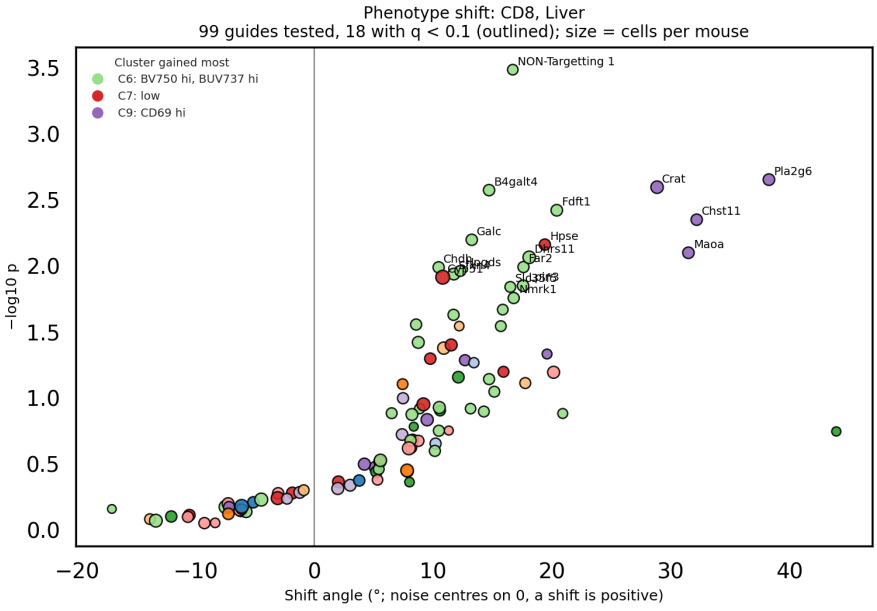
SECTION 6

Phenotype shift: do a guide's cells become a different kind of cell?

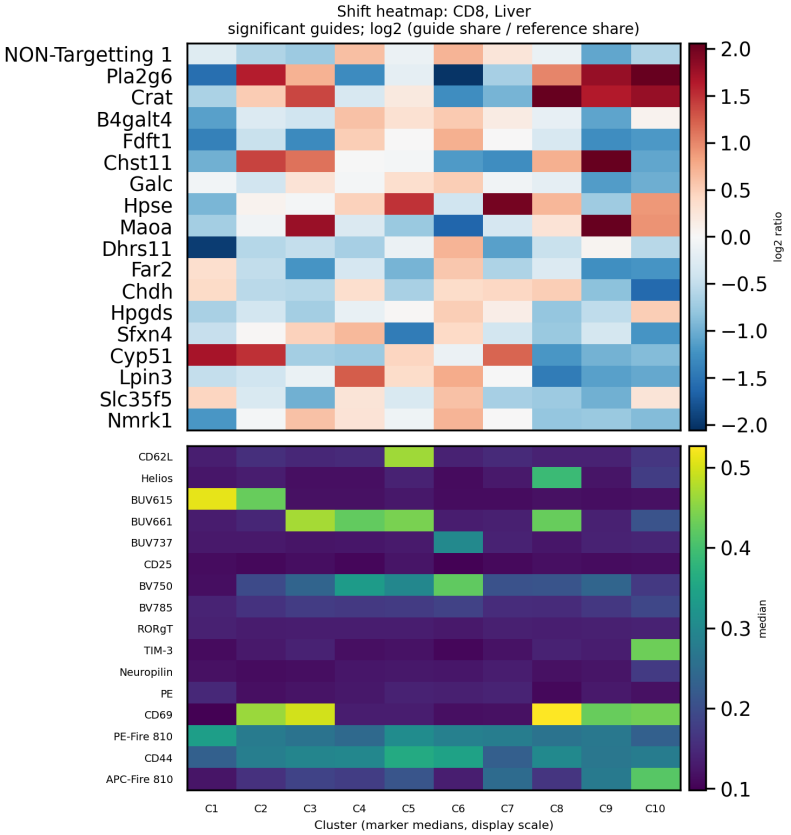
The barcoded cells of each population are clustered (FlowSOM), and each guide in each mouse gets a fingerprint: its share of cells in each cluster. That is compared with what its tags predict in the same mouse. One mouse is too noisy to show a shift, but noise points a different way in each mouse while a real shift points the same way in all of them, so the test asks whether the mice agree. The **shift angle** is its size: 0° for none, 90° for cells in entirely different clusters.



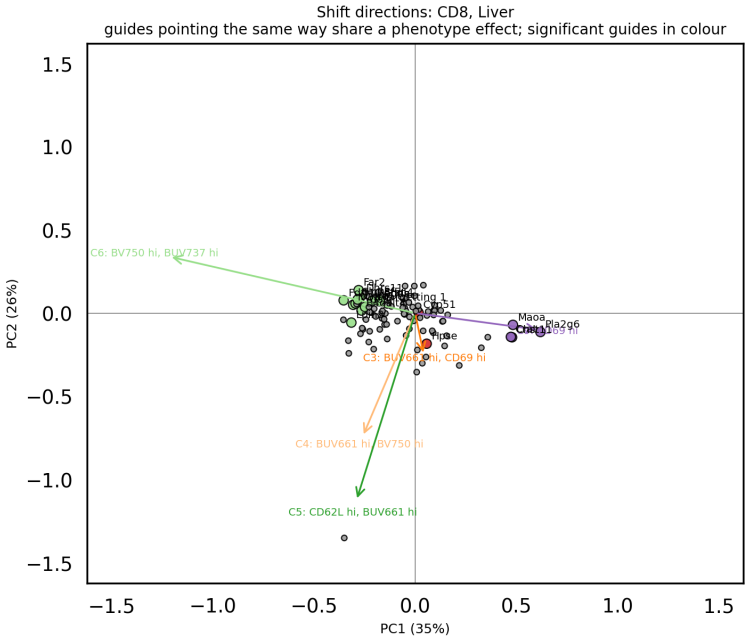
Phenotype shift tab: settings on the left (here 10 clusters, tag-model reference, tag spillover corrected before clustering, immune states on), run summary on the right: 1,685 guide tests in 3 populations, 18 significant (q < 0.1).



Shift plot, CD8 T cells in the liver: shift angle against −log10 p, coloured by the cluster each guide gains most. All 18 significant shifts of the run are in this group.



Shift heatmap, CD8 liver: per significant guide, log2 of its cluster shares over the reference (top), and each cluster's marker medians (bottom).

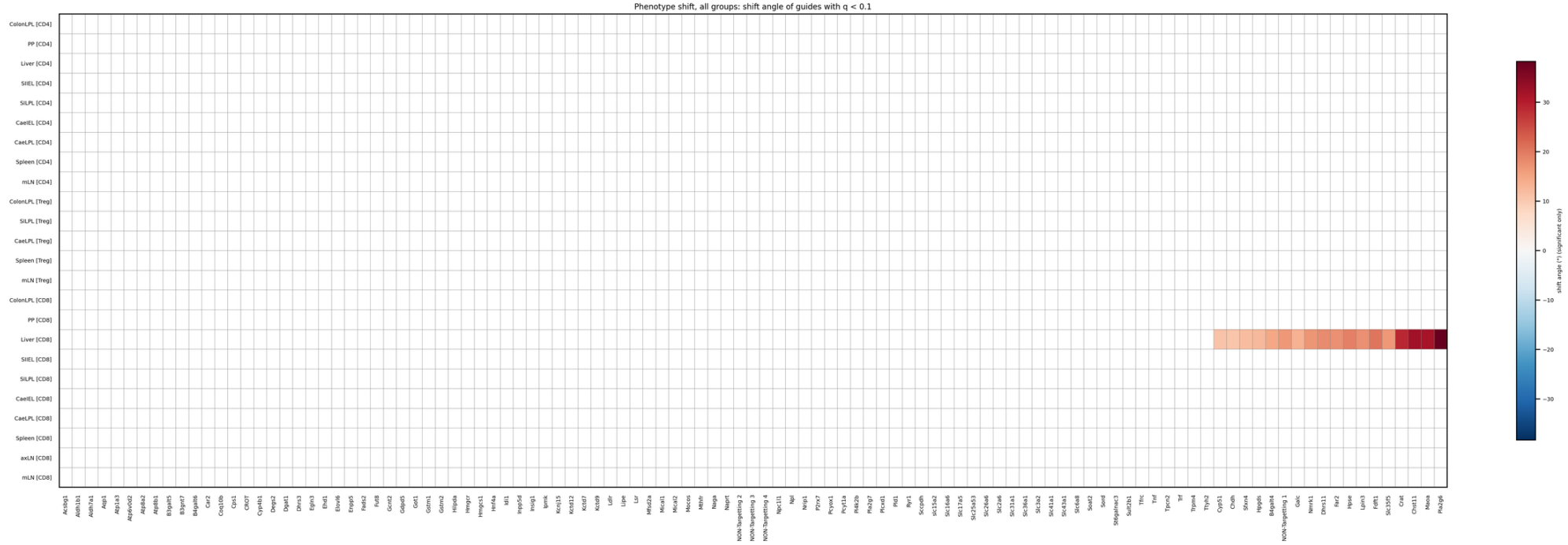


Direction map: guides pointing the same way share a phenotype effect; arrows show which cluster each direction moves cells into.

What to check

- Here the top hit is a non-targeting guide (NON-Targetting 1), and most shifted guides gain cluster C6, defined by BV750 and BUV737, channels with no marker assigned on this panel. Both point to something other than the knockouts: run **Check calibration**, and look at which channels the clustering uses (*Channels* in the settings) before interpreting these guides.
- Groups reported as "no guide had 10 cells in 3 mice" are not tested (here, for example, colon IEL in all three populations): too few barcoded cells per mouse.

Every tissue at once

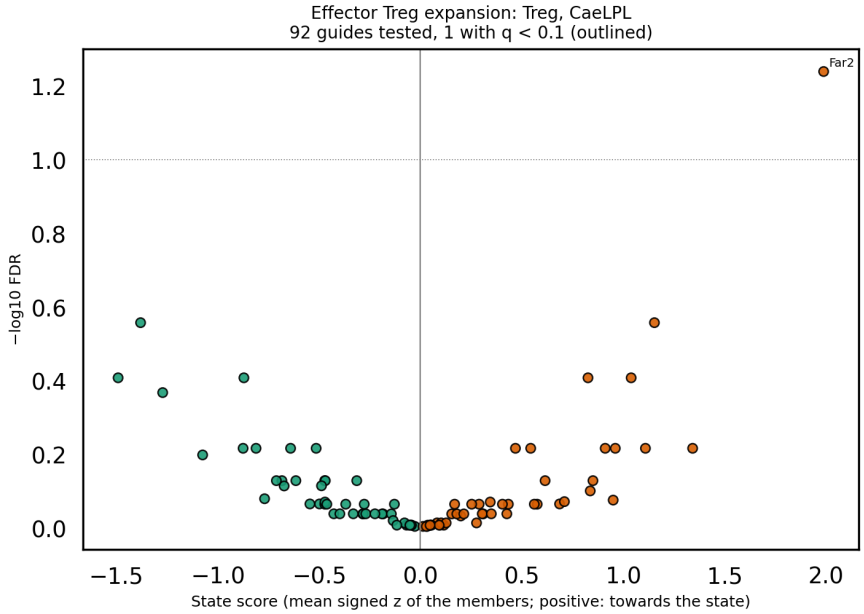


Summary heatmap: the shift angle of every guide (columns) in every group and population (rows), coloured only where significant. A guide shifting in several tissues would show as a column of colour; here only the liver CD8 row is coloured.

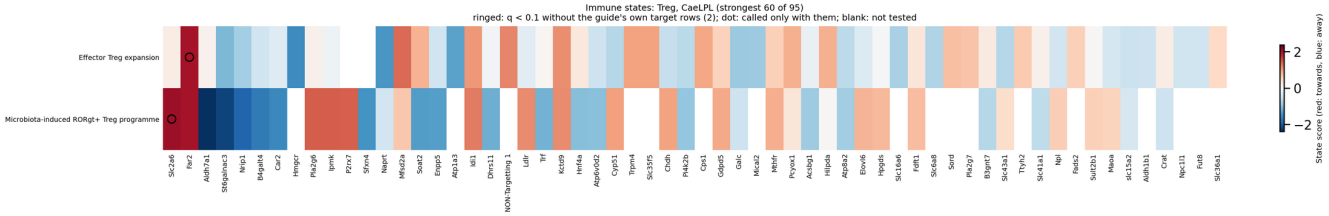
SECTION 6

Immune states: what kind of change?

The shift says that a guide's cells have changed kind; the immune states say which change, in the terms of the Flow Set Enrichment library. An immune state is a short list of expected changes, for example *effector Treg expansion*: more CD44+, ICOS+, KLRG1+ and CD103+ cells among Tregs, fewer CD62L+. Only the items the panel can measure are used, and each guide is scored with and without the items of its own target gene.



State volcano, effector Treg expansion in caecum LPL Tregs: 92 guides tested; Far2 knockouts move towards the state (q < 0.1).



Immune-state heatmap, caecum LPL Tregs: states (rows) against guides (columns), red towards the state and blue away; ringed where significant (Far2 for effector Treg expansion, Slc2a6 for the RORγt+ Treg programme). 759 guide × state tests in the run, 4 significant.

What to check

- A state needs at least 3 measurable items on the panel; the *States* view lists which rows the panel can and cannot score.
- A call that holds only with the guide's own target items (a dot in the heatmap) is the marker result, not a state change.

SECTION 6

Export analysis

Export analysis writes one folder named after the populations: the counts and percentages (also as an R-style pct_matrix.csv), QC, samples, results, library sizes, clones, marker results, the pathway, gene-group, phenotype-shift and immune-state tables, settings.txt, a PNG per figure and one PDF report. For this run that is 364 files. Statistics that are out of date cannot be exported: run them again first.

Checklist before you trust a result

1–2	Backbone is the FRET control; matched unstained similarity above about 0.9; tags and libraries right.
3	Thresholds in the gaps of every tag; Cells gate holds the barcoded cloud.
4	All combinations present; after correction, barcoded and untransduced cells line up off-tag.
5	Tandem and calibration lines read sensibly; no greyed-out samples you need.
6 Data	Enough barcoded cells per sample; expanded clones understood.
6 Groups	Pairing by mouse; the acquisition date in the model when days differ; groups spread across days.
6 Results	Non-targeting guides are not hits; hits hold in leave one out; Counts and Frequencies compared.
6 Shift	Check calibration run; clustering channels are markers; non-targeting guides not shifted.

Where to read more: the plugin README (settings and mechanics of every step), the install guide (installing, updating, cohorts over several days), and each section's Help button in the app.